Adish Hussain

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Education

VIT Bhopal University, Bhopal — 2023–2027
B.Tech in Computer Science & Engineering (AI & ML) — CGPA: 7.8/10
Relevant Coursework: DSA, Operating Systems, Computer Networks, DBMS, Machine Learning, OOP

Technical Skills

Programming: Python, C++, SQL
Backend: Express.js ,Fast API, Flask, REST APIs
ML/AI: TensorFlow, Scikit-learn, Keras, NumPy, Pandas
Databases: MySQL, PostgreSQL,Vector Database
Enterprise/Other: SDLC, Git, Docker, AWS, Jupyter Notebook, VS Code

Experience

IT Intern – Project Neogenisis — May 2026 – Jun 2026
Sun Pharmaceutical Industries Ltd. (R&D-IT), Gurugram

- Resolved functional and build-level issues during the Neogenisis software rollout by debugging across lab systems and test cycles.
- Executed software lifecycle upgrades across 10-15 R&D lab systems, deploying updated releases and deprecating legacy versions per SOP compliance standards.
- Designed and tested internal APIs while applying enterprise SDLC and change-management practices within an SAP-integrated environment.

Software Development Intern — Jun 2026 – Aug 2026
Sopra Steria India Ltd., Noida

- Built an AI-powered OData service orchestration platform (FastAPI, Neo4j, ChromaDB, Docker Compose) enabling natural-language querying across 200+ enterprise SAP OData entities.
- Designed an LLM-based reasoning layer with query-plan caching and vector memory/RAG, reducing repeated-query latency by 90% and achieving 90% query-plan accuracy.
- Validated the platform against real Sopra Steria SAP service data and public Northwind OData services, implementing metadata discovery, query planning, pagination, and cross-service joins.
- Layered in 17 ML algorithms for automated data analysis, plus RBAC, JWT authentication, audit logging, and n8n-based workflow sharing (Email, WhatsApp, Slack).

Projects

Deep Packet Inspection (DPI) Engine | *C++* — GitHub

- Engineered a C++ network traffic analyzer parsing PCAP files with deep packet inspection by decoding Ethernet, IP, and TCP/UDP headers at the byte level, processing 10,000+ packets/sec.
- Implemented TLS SNI extraction and five-tuple flow hashing to classify encrypted application traffic from YouTube, Facebook, and Google with 85–92% classification accuracy.
- Designed a multi-threaded processing pipeline and stateful flow-tracking system for session reconstruction, reducing per-packet inspection latency to 0.5–1 ms across 10,000+ concurrent flows.

Vector Database with RAG Pipeline | *C++, Docker, Ollama* — GitHub

- Built a vector database in C++ implementing HNSW (O(log N)), KD-Tree, and Brute Force search across Cosine, Euclidean, and Manhattan distance metrics, indexing 10,000+ vectors with 5 ms average query latency.
- Implemented HNSW — the same algorithm used by Pinecone, Weaviate, and Chroma — with multilayer graph construction, beam search, and bidirectional neighbor linking, achieving 5× speedup over brute-force search.
- Developed a RAG pipeline integrating `nomic-embed-text` (768D embeddings) and `llama3.2` for document ingestion, semantic retrieval, and context-grounded LLM responses.
- Exposed a full REST API with CRUD, benchmark, and RAG endpoints across 12+ routes; containerized the entire stack using Docker Compose with health-check orchestration.

AI-Powered Enterprise OData Orchestration Platform | *Python, FastAPI, Neo4j, ChromaDB, Docker* GitHub

- Built an AI-assisted OData orchestration platform using FastAPI, Neo4j, ChromaDB, Docker, and a JavaScript frontend, enabling natural-language querying across 200+ enterprise OData entities.
- Tested against real Sopra Steria service data and public Northwind OData services, supporting metadata discovery, query planning, pagination, and joins with 90% query-plan accuracy; built a Pytest-based automated test suite to validate query-plan correctness and API endpoint reliability.
- Designed an LLM-based reasoning layer with query-plan caching and vector memory/RAG, reducing repeated-query latency by 90%; added RBAC, audit logging, guarded CRUD workflows, and ML analytics.

Certifications

- Machine Learning Specialization – Andrew Ng, DeepLearning.AI — *Coursera, 2024*
- Python for Data Science – Oracle Cloud Infrastructure — *Oracle, 2025*
- Introduction to Data Analytics — *Coursera, 2025*